



THE UNIVERSITY OF
NOTRE DAME
A U S T R A L I A

MED 4000
WEEK 4.05

TUTOR GUIDE

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The cases and materials in this guide were reviewed by;

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RELEVANT PRIOR LEARNING

MED1000

- 1.20 Gas Exchange & Shunt (L)
- 1.21 Respiratory Mechanics (ER)

MED2000

- 2.11 Mechanics of breathing I (L)
- 2.12 Mechanics of breathing II (L)
- 2.12 Regulation of breathing (L)
- 2.13 Mechanics of breathing III (L)
- 2.13 Gas Exchange & transport (L)

MED 3000

- 3.08 Neonatal paediatrics_newborn breathing problems [SC tutorial]
- 3.08 Neonatal problems_low birth weight, prematurity and jaundice [SC tutorial]

WEEK 05_ INTENSIVE CARE MEDICINE		
Yr	Wk	LEARNING OBJECTIVES
4	5	- Define Acute Respiratory Failure - Know the causes of Non-Hypercapnic respiratory failure and explain the underlying pathophysiology - Know the principles of management of Non-Hypercapnic respiratory failure. - Describe the basic modes of mechanical ventilation and explain the process for commencing a patient on basic mechanical ventilation. - Know the key clinical parameters that are monitored on a ventilated patient and be able to interpret changes in the clinical parameter and plan the initial management. - Outline the key clinical features and presentation of acute respiratory distress syndrome in adults. - Outline the principles of assessment and management of acute respiratory distress syndrome in adults.
4	5	- Know the causes of Hypercapnic respiratory failure and explain the underlying pathophysiology. - Know the principles of management of Hypercapnic respiratory failure. - Outline the principles of non-invasive ventilation and how it is implemented in the clinical setting. - Outline the ongoing assessment and management plan with Hypercapnic respiratory failure commenced on non-invasive ventilation.
4	5	- Review the assessment and management of acute severe asthma and be able to explain the underlying pathophysiology. - Know the medications used in the management of acute severe asthma, including Salbutamol, Ipratropium Bromide, Adrenaline, Magnesium, Hydrocortisone, and Aminophylline - Be able to outline a management plan for a patient. - Outline the complications of acute severe asthma.
4	5	- Revise the components of the Apgar scoring system and be able to assess a newborn baby. - Know the key clinical parameters in the assessment of the newborn infant and be able to implement appropriate resuscitation measures based on these clinical parameters. - Summarise the key difference in neonatal and adult physiology in the respiratory system, cardiovascular system, fluid & electrolyte metabolism, immune system and thermoregulation. - Revise the assessment and management of respiratory distress syndrome in newborns and outline the key clinical parameters that would require transfer to a neonatal intensive care unit.

CASE ONE

Short case number: 4_05_01

Category: Ventilation

Discipline: Intensive Care Medicine

Setting: Intensive Care Unit_urban

Topic: Respiratory Failure_Community Acquired Pneumonia

CASE

You are the Resident Medical Officer on rotation in the Intensive Care Unit.

Your Registrar informs you she has accepted an admission into the ICU from the Emergency Department who will be arriving shortly. She would like you to undertake the admission of the patient. She hands you a copy of the Emergency Admission Note which is as follows:

PC: 51 year old male presents with Severe Shortness of Breath

BG: HTN

ALLERGIES: Nil known

MEDICATIONS: Perindopril 5mg Daily

SOCIAL: Non Smoker

Social ETOH 20gm/week

Firefighter

HPC: Presents with Fevers, cough, and increasing Shortness of Breath for 5 days

- Increasing over this time
- Exercise tolerance deteriorated from being able to walk >2km without SOB to Shortness of Breath at rest
- Associated with intermittent fevers for the last 5 days up to 40 degrees
- Dry cough
- Lethargy 4 days
- Dizziness on standing for the last 1 day

Associated History

- nil recent travel
- nil contacts unwell
- nil previous episodes of similar illness

Deteriorated rapidly this afternoon

- Increased SOB
- Wife called ambulance

CASE

On ambulance arrival at patients home

- HR 126, BP 86/50, RR 38, SaO₂ 78% on RA, mild central cyanosis present
- Placed on oxygen 15L Non re-breather mask, with Saturations up to 91%

ON EXAMINATION:

AIRWAY:

Assessment: Patent and Protected on arrival

BREATHING:

Assessment:

RR 40, SaO₂ 90% on 15L NRB Oxygen Mask

Nil central cyanosis

Trachea Midline

Markedly increased work of breathing with accessory muscle use

Decreased air entry bilaterally, crackles and crepitations throughout both lung fields

Investigations:

CXR – marked bilateral infiltrate throughout all lung fields, fluid in fissure

ABG – pH 7.4, PCO₂ 40.1, PO₂ 52.2, HCO₃⁻ 27, BE -1.3 (on 15L Oxygen)



CASE

Management:

Intubation and Ventilation

Rapid Sequence Induction performed

Intubated first pass ETT Size 8.0, Grade 1 View

ETT placement confirmed via auscultation, CO2 monitoring, and CXR

Placed on Ventilator

CIRCULATION:

Assessment:

HR 130/min regular, BP 100/55, CR = 2.5 seconds, cool peripheries

Investigations:

Bloods sent for EUC/LFT/CMP/FBC/CRP

Blood Cultures sent

Management:

IVC x2 sited (18G)

IVF bolus given – 1x 1L Normal Saline

Blood pressure 110/58 post bolus

DISABILITY

Assessment:

GCS 13/15 (E3V4M6) (Prior to Intubation)

BSL 7.3

Management:

Morphine and Midazolam infusion commenced on Intubation

EXPOSURE:

Temperature 38.1 on arrival

Nil obvious Injuries on complete exposure

IMPRESSION:

1. Severe Bilateral Community Acquired Pneumonia
2. Sepsis secondary to Pneumonia
3. ? Early Acute Respiratory Distress Syndrome

PLAN:

Ventilation

Intravenous Antibiotics

IDC

Strict fluid balance

Intensive Care Registrar contacted, will kindly review ASAP re: admission

CASE
Discussed with Respiratory Consultant, happy to accept care of patient

QUESTIONS
1. What are the key features of the clinical presentation of Acute Respiratory Failure? 2. Outline the causes and underlying pathophysiology of Non-Hypercapneic (also known as Type I) Respiratory Failure. 3. What are the principles of management of Respiratory Failure in this patient? 4. Outline the basic modes of mechanical ventilation. 5. The Nurse asks you what settings you would like to dial up on the ventilator. Outline the initial ventilator settings for this patient 6. Describe how you would assess the clinical progress of this patient. 7. What antibiotics, including dosage and route of administration, are indicated in the above patient given the information? 8. Briefly outline the Acute Respiratory Distress Syndrome.

Resources:

- Berston, AD, & Soni, N (2009). Oh's intensive care manual (6th ed.). Philadelphia: Elsevier.

Other Suggested Resources:

- Kaynar, AM (2010). Respiratory Failure. Retrieved from:
<http://emedicine.medscape.com/article/167981-overview>
- Therapeutic Guidelines. Antibiotic Guidelines.

ANSWERS

1. What are the key features of the clinical presentation of Acute Respiratory Failure?

Respiratory Failure is failure of the respiratory system in one or both of its gas exchange functions, which results in an inability to meet the metabolic demands of the body. The two major functions of the respiratory system are to provide an interface whereby oxygen and carbon dioxide can be exchanged, oxygen from inspired air into the blood, and carbon dioxide elimination from the blood into the alveolar air to be expired.

Definitions:

Hypoxemic Respiratory Failure

$\text{PaO}_2 < 60\text{mmHg}$ (with FiO_2 21%)

Normal or low PaCO_2

[Also known as Type 1 Respiratory Failure]

Hypercapnoeic Respiratory Failure

$\text{PaCO}_2 > 50\text{mm Hg}$

Normal or low PaO_2

[Also known as Type 2 Respiratory Failure]

2. Outline the causes and underlying pathophysiology of Non-Hypercapneic (also known as Type I) Respiratory Failure.

Pathophysiologic Causes of Acute Respiratory Failure

Low inspired partial pressure of oxygen

- High altitude

Hypoventilation

- Respiratory centre depression: i.e. drug overdose, head injury, encephalopathy, anaesthesia
- Neuromuscular Dysfunction: spinal cord injury, motor neurone disease, Guillain-Barre syndrome, paralyzing agents, myasthenia gravis
- Chest wall abnormalities: pleural fibrosis, kyphosis
- Muscular Dysfunction: myopathy, muscular dystrophy, fatigue

Ventilation perfusion mismatch [most common cause of hypoxemia]

Part of lung ventilated but not perfused (Shunt) AND/OR perfused but not ventilated

- Pneumonia
- Pulmonary oedema
- Atelectasis
- Pulmonary contusion / haemorrhage
- Cardiac defect / Shunt
- Pneumothorax

3. What are the principles of management of Respiratory Failure in this patient?

Oxygen supplementation

- Maintain PaO₂ > 60mmHg and/or SaO₂ >90%
- increased inspired oxygen concentration
- increased CPAP/PEEP to maintain recruitment of collapsed lung, increased Functional residual capacity and minimise intrapulmonary shunt

Ventilatory Support

- Ventilatory demand exceeds capacity due to shunt and V/Q mismatch
- Need to provide support in the form of mechanical ventilation
- Most likely will need invasive mechanical ventilation due to fatigue and extensive loss of gas exchange capacity due to bilateral infiltrative pneumonia

Diagnose and Treat Cause of Respiratory Failure

- appropriate antibiotics

4. Outline the basic modes of mechanical ventilation (Copy and paste from emedicine)

Assist-control ventilation

Volume Control

- Set Tidal volume and Respiratory rate
- Ventilator has mandatory number of breaths per minute set, in which it delivers the set tidal volume. Patient able to take spontaneous breaths but not synchronised with ventilator initiated breaths, can lead to “breath stacking”, therefore often need sedation and/or paralysis to achieve patient-ventilator synchrony

Pressure Control

- Set inspiratory pressure and respiratory rate
- Ventilator has mandatory number of breaths per minute set, in which it delivers the set inspiratory pressure.

Pressure Support Ventilation

- Set inspiratory pressure only
- Patient initiates breath and machine delivers set level of inspiratory pressure
- Cannot use in paralysed patients / heavily sedated / apnoeic
- Pressure support of 3.5 – 14.5cmH₂O needed to overcome the work of breathing added due to the Endotracheal tube

Synchronised Intermittant Mandatory Ventilation

- set respiratory rate and inspiratory pressure
- Patient breaths are synchronised with ventilator mandatory breaths, leading to less breath stacking and improved patient comfort

5. Briefly outline the important effects of positive pressure invasive mechanical ventilation on the cardiovascular system
--

Decreased Cardiac output due to decreased venous return

6. The Nurse asks you what settings you would like to dial up on the ventilator. Outline the initial ventilator settings for this patient

For a simple mechanical ventilator:

Intermittant Mandatory Ventilation: Patient Paralysed/Sedation [No spontaneous effort]

Volume Control Ventilation

Fraction inspired oxygen (FiO₂) – initially 100% then wean to maintain SaO₂ 90 - 94% and/or PaO₂ 60mmHg – 90mmHg

Respiratory Rate – 12-14 (higher rate for patient with increased metabolic demand/severe metabolic acidosis (require higher minute ventilation therefore higher RR)

Tidal Volume – 5-8mls/kg

PEEP – 5cm H₂O

Airway Pressure [Limit alarm] – 35mmHg

Pressure Control Ventilation

Airway Pressure Alarm – 30mmHg

PEEP – 5cmH₂O

Inspiratory Pressure – adjust to achieve desired tidal volume, normally between 5-15mmHg

FiO₂ – 100% initially then wean as above

Respiratory rate

7. Describe how you would assess the clinical progress of this patient.

Physical Examination

Monitoring trends

- SaO₂

Ventilatory Requirements

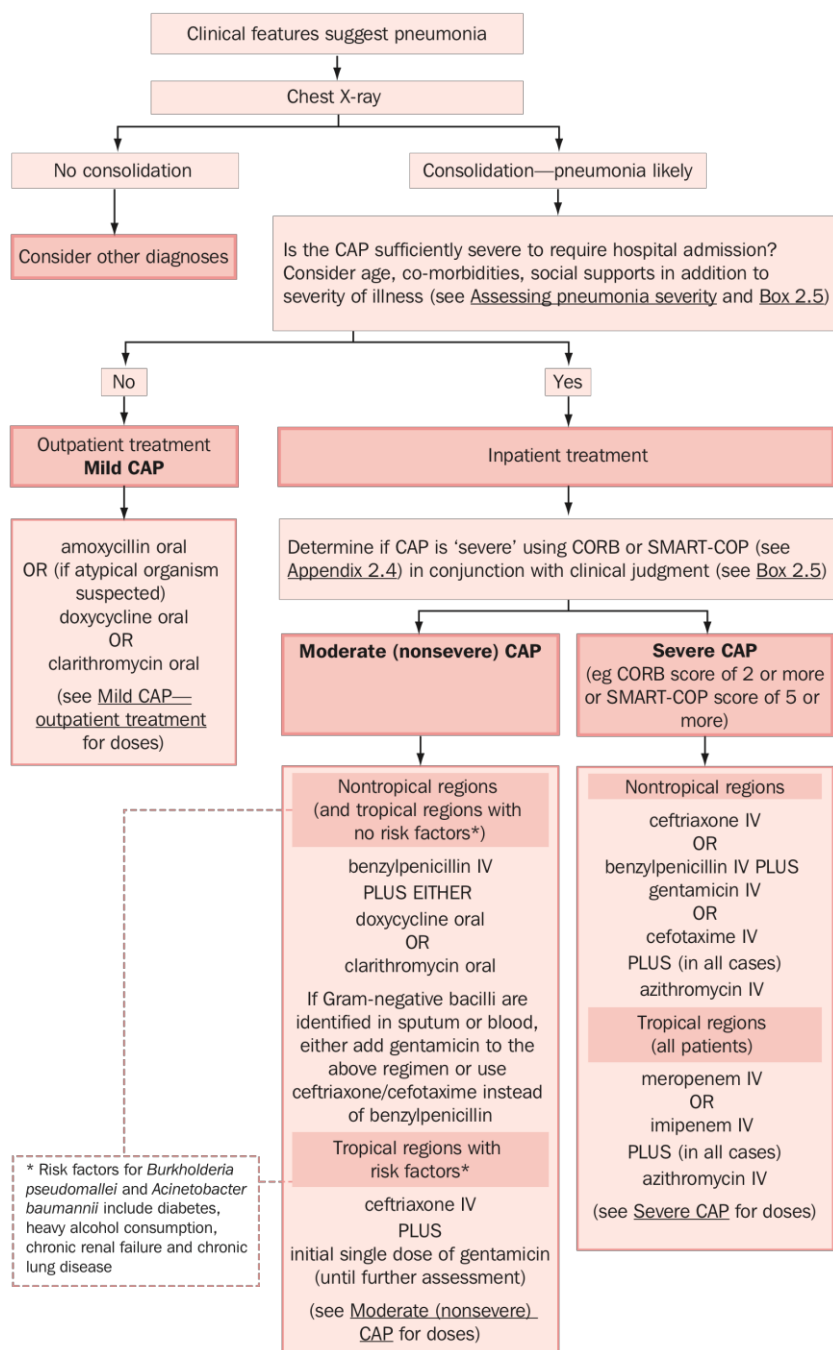
- FiO₂
- Inspiratory pressure

Arterial Blood Gases

- PaO₂
- PaCO₂
- pH

CXR

8. What antibiotics, including dosage and route of administration, are indicated in the above patient given the information?



9. Briefly outline the Acute Respiratory Distress Syndrome.

Diffuse inflammatory process, characterised by widespread inflammation in the lungs, causing hypoxemic respiratory failure and diffuse alveolar damage.

Predisposing Conditions: Table 16.1: Predisposing conditions for ARDS

Conditions associated with Systemic Inflammatory Response

- Severe sepsis
- Extensive Tissue Injury (i.e. Severe trauma)
- Cardiopulmonary Bypass

Local Lung Conditions

- Aspiration
- Pneumonia
- Pulmonary Contusion
- Ventilator induced lung injury

Embolism Syndromes

- Fat
- Amniotic

Miscellaneous

- Blood transfusions
- Drugs
- Pancreatitis
- Intracranial hypertension

From: Marino (2009). The little ICU book of facts and figures.

Clinical Manifestations

Tachypnoea

Progressive hypoxemia

Fever

Bilateral pulmonary infiltrates

Diagnostic Criteria: Table 16.2 Diagnostic Criteria for ARDS and ALI

Acute onset

Presence of predisposing condition

Bilateral infiltrates on CXR

PaO₂/FiO₂ <200mmHg for ARDS

Pulmonary artery occlusion pressure <18mmHg OR no clinical evidence of left heart failure

Principles of Management

Mechanical ventilation

Moderate to High PEEP (5-10) + low tidal volumes (6ml/kg predicted body weight)

End inspiratory Plateau pressures <30cmH₂O

Aim for PaO₂ >60mmHg

Permissive hypercapnea

CASE TWO

Short case number: 4_05_02

Category: Ventilation

Discipline: Intensive Care Medicine

Setting: Lithgow Hospital

Topic: Type II Respiratory Failure

CASE

You are the Career Medical Officer (CMO) on duty in the Lithgow High Dependency Unit (HDU). You receive a call from the Emergency CMO to request a review of a patient in the Emergency Department who has presented with Shortness of Breath on a background of Chronic Obstructive Pulmonary Disease (COPD), secondary to a 50 pack year history of smoking. The CMO informs you that this patient is currently receiving Non Invasive Ventilation (NIVV) via a BiPAP ventilation system.

In the Emergency Department, you review the patient notes prior to seeing the patient:

PC: 75 year old female presents with Shortness of Breath

BG:

COPD

- 40 + pack years smoking
- Ceased smoking 2 years ago
- 3x admissions to Hospital for Infective Exacerbations of COPD
- Nil previous admissions to HDU/ICU
- Not on home oxygen
- Exercise tolerance 20m

Hypertension

Hypercholesterolemia

Allergies: Penicillin – ANAPHYLAXIS

Medications:

Ramipril 5mg Daily

Seretide 250/25 2 puffs daily

Salbutamol 2-6 puffs PRN

Spiriva 18mcg Inhalation daily

Simvastatin 40mg Nocte

Social:

Lives with husband, independent with Activities of Daily Living

Nil ETOH

Nil mobility aids; 2 stairs at front of the house

CASE

HPC:

4 day history of increasing shortness of breath

- Gradual increase
- worse on exertion
- now occurring at rest

Associated with increased sputum production, yellow in colour

Mild fever last evening, 38.1 degrees

Exercise tolerance now <5m (previously 20m)

Seen by GP yesterday

- commenced on Prednisone 50mg Daily
- Nil obvious improvement

OE:

A: Patent and Protected

B: RR 32, SaO₂ 86% on Room Air. Marked increased work of breathing, purse lipped. Nil central cyanosis. Equal air entry, globally decreased air entry. Mild end expiratory wheeze, scattered crepitations.

Commenced on Non invasive ventilation with Inspired Oxygen of 30%. SaO₂ improved to 94%, work of breathing slowly decreasing.

C: HR 98, BP 154/81, CR = 2.5 seconds, mildly dry lips. HSDNM. JVPNR. Nil peripheral oedema.

IVC inserted. Slow IVF rehydration commenced with Normal Saline at 80mls/hr.

D: GCS 15/15, PEARL. Moving all limbs equally. BSL 7.1

E: Temperature 37.8

INVESTIGATIONS:

Bloods – EUC, LFT, CMP, FBC, Blood Cultures all sent

CXR – Hyperinflated, nil obvious consolidation

ABG – pH 7.310, PCO₂ 65.7, PO₂ 62.1, HCO₃⁻ 32, BE 4.0 (on 30% FiO₂)

CASE



IMPRESSION:

1. Infective Exacerbation of COPD

PLAN:

Continue on NIVV

Antibiotics

HDU CMO contacted for review re: admission to HDU for continued NIVV

QUESTIONS

1. What are the clinical features of Hypercapneic (Type II) Respiratory Failure?
2. Outline the causes and underlying pathophysiology of Hypercapneic Respiratory Failure.
3. What are the principles of management of Hypercapneic Respiratory Failure?
4. Describe the Principles of Non Invasive Ventilation.
5. Outline how you would set up a Non Invasive Ventilator for the above patient.
6. Describe how you would assess the clinical progress of this patient.
7. What antibiotics, including dosage and route of administration, are indicated in the above patient given the above information?
8. Outline the management plan for the above patient, including considerations in the plan for further treatment should the clinical condition deteriorate.

Resources:

- Berston, AD, & Soni, N (2009). Oh's intensive care manual (6th ed.). Philadelphia: Elsevier.

Other suggested Resources:

- Kaynar, AM (2010). Respiratory Failure. Retrieved from:
<http://emedicine.medscape.com/article/167981-overview>
- Soo Hoo, GW (2010). Ventilation, Noninvasive. Retrieved from:
<http://emedicine.medscape.com/article/304235-overview>
- Therapeutic Guidelines. Antibiotic Guidelines.

ANSWERS

1. What are the clinical features of Hypercapneic (Type II) Respiratory Failure?

Hypercapnoeic Respiratory Failure

PaCO₂ > 50mm Hg

Normal or low PaO₂

[Also known as Type 2 Respiratory Failure]

Precipitants (Oh's Intensive Care Manual)

Infection 50%

Heart failure 35%

Retained secretions/coexistent medical problems/PE/medications 25%

Uncontrolled oxygen administration

2. Outline the causes and underlying patho-physiology of Hypercapneic Respiratory Failure.

Causes:

Respiratory Centre Depression

Drug Overdose – Narcotics, Benzodiazepines

Hypoventilation – secondary to brain injury/herniation

Airway Disorders

Asthma

Chronic Obstructive Airways Disease (COAD)

Chest Wall Abnormalities

Neuromuscular Disorders

Muscular Dystrophy

Spinal Muscular Atrophy (SMA)

Pathophysiology:

At a constant rate of carbon dioxide production, PaCO₂ is determined by the level of alveolar ventilation (V_a) in which VCO₂ is ventilation of carbon dioxide and K is a constant value (0.863)

$$(V_a = K \times VCO_2) / PaCO_2$$

A decreased in alveolar ventilation can result in overall (minute) ventilation or an increase in the proportion of dead space ventilation. A reduction in minute ventilation is observed primarily in the setting of neuromuscular disorders and CNS depression. In pure hypercapnic respiratory failure, the hypoxemia is easily corrected with oxygen therapy.

3. What are the principles of management of Hypercapneic Respiratory Failure?

Ensure adequate Oxygenation

Manage respiratory acidosis

- may require ventilation to help lower PaCO₂
- Can be invasive (ETT) or non-invasive (CPAP/BIPAP)

Unload Respiratory Muscles

Minimize development of hyperinflation

4. Describe the Principles of Non Invasive Ventilation.

Positive pressure ventilatory support without intubation

Delivered via face or nasal mask

Goals of Therapy in COPD Exacerbations

Unload respiratory muscles

Augment ventilation and oxygenation

Reduce CO₂ and correct acidosis until underlying problem can be reversed

Offset the adverse effects of sleep or position-induced adverse changes to ventilation

Indications NIV in COPD

Patient co-operation

Acute dyspnoea

Increased Work of breathing

Respiratory Rate >24-28/min

PaCO₂ >45mmHG with pH <7.35 despite optimal medical treatment

Contraindications

Decreased level of consciousness / inability to protect airway

Presence of indicators for intubation and mechanical ventilation (in patients suitable for intubation/ventilation)

Upper GIT bleeding / Intractable emesis

Side Effects

Patient discomfort

Skin necrosis

Gastric distention & aspiration

5. Outline how you would set up a Non Invasive Ventilator for the above patient.

Settings

Aim to achieve adequate tidal volumes of 5-7ml/kg

Aim to reduce respiratory rate to <25/min

Inspired Oxygen

- lowest possible oxygen to achieve PaO₂ >60mmHg or SaO₂ >88-90%

Inspiratory positive airway pressure (IPAP)

- start at 10cmH2O
- increase by 2cmH2O if persistent hypercapnia
- maximum 20-25cmH2O

Expiratory positive airway pressure (EPAP)

- synonymous with PEEP
- start at 5cmH2O and titrate if need to improve oxygenation
- maximal EPAP 10-15cmH2O

Pressure Support

- difference between IPAP and EPAP
- increasing pressure support will theoretically increase tidal volume, and thus contribute to decreasing PaCO2 ($MV = TV \times RR$)

Back up respiratory rate

- set at 8-12/min

6. Describe how you would assess the clinical progress of this patient.**Observations**

- increased SaO2, decreased RR

Clinical Examination

- decreased WOB

Investigations

- ABG, particularly looking at PaCO2 decreases, PaO2 increases, normalisation of pH

Predictors of Success with Trial of NIVV (1-2hrs)

- decrease in PaCO2 >8mmHg
- Improvement in pH >0.06
- Correction of respiratory acidosis

Predictors of Failure

- Acidosis with pH <7.25
- Hypercapnoea PaCO2 >80mmHg
- APACHE II score >20
- Decreased LOC, with GCS <8
- Failure of improvement within 12-24hrs of NIVV

7. What antibiotics, including dosage and route of administration, are indicated in the above patient given the above information?

8. Outline the management plan for the above patient, including considerations in the plan for further treatment should the clinical condition deteriorate.

Set Goals and Limits for treatment

- often a medical + patient decision, i.e. suitable/wants NIVV, not suitable or patient refuses intubation/ventilation
- if suitable for intubation and ventilation, set parameters at beginning of NIVV, i.e. if deteriorates to pH <7.2, RR increases to 30 etc will require intubation
- consider the patients baseline level of physiology and functioning, i.e. no point in aiming for normocarbica when patient's baseline PaCO₂ is 50mmHg (Chronic compensated respiratory acidosis). Set goals accordingly, i.e. aim PaCO₂ <55mmHg, SaO₂ >90%, PaO₂ >60mmHg, and titrate ventilation to try to achieve these parameters

Consider precipitants/underlying causes

- i.e. infective exacerbations of COPD will require IVAB, chest physiotherapy

Involve the patient, family and treating physicians with the treatment planning and decision making process

- family and treating physicians will know patients baseline level of functioning, physiology and patient wishes, and can help to guide treatment accordingly

If patient refuses Intubation, or it is not Medically Indicated/Appropriate

- will require treatment limitations to be documented, including not for resuscitation status

CASE THREE

Short case number: 4_05_03

Category: Ventilation

Discipline: Intensive Care Medicine / Emergency Medicine

Setting: Ballarat Hospital

Topic: Acute Severe Asthma

CASE

You are the Junior Doctor on Evening shift at Ballarat Hospital Emergency Department. The Triage Nurse informs you that a 48 year old female is being transferred to the Resuscitation Bay from the Ambulance Bay. She is severely short of breath.

You arrive in the Resuscitation Bay as the Ambulance officers are transferring the patient onto the bed. You can see she is extremely short of breath. You ask the Ambulance Officers for a handover.

The Ambulance Officer states they were called to the patient's home as she had become very short of breath. Her son was at home, and he stated that she had had a cold for the last 3 days, and that it had been making her asthma play up. She had been using her Ventolin puffer "a lot" but today it wasn't helping. He called the ambulance when she stopped being able to talk to him because she was so breathless. She hasn't been to the local doctor for this illness. The son stated that his mother has been in hospital for her asthma before, about 6 months ago, and at that time she needed to go to ICU she was so sick.

The Ambulance officers state she is a smoker, and has no known allergies, but have no further background or history.

On ambulance arrival her observations were:
HR 120, BP 156/70, RR 40, SaO₂ 88% on Room Air

The ambulance officers commenced the patient on Oxygen, and have given her 5mg Salbutamol via nebuliser.

You commence your ABCDE Assessment and Management of this seriously unwell patient:

A

Assessment: Nil obvious lip/face/neck swelling, nil stridor.

Management: Nil currently

B

Assessment: RR 38, SaO₂ 93% on 15L oxygen. Speaking in single words only. Marked increased accessory muscle use, tripod posture. Trachea midline. Air entry equal but markedly decreased bilaterally, expiratory wheeze audible.

ABG: pH 7.292, PCO₂ 64.8, PO₂ 90.2, HCO₃⁻ 31, BE 1.7

CXR: Hyperinflated, nil evidence of Pneumothorax.

CASE

Management: Oxygen. Continuous Nebulised Salbutamol. Atrovent. Hydrocortisone 200mg IV.



C

Assessment: HR 116, BP 147/72, CR = 2.5 sec. HSDNM. Nil peripheral oedema.

Management: IVC x2 sited peripherally (18G). Bloods sent for EUC/FBC/LFT/CMP.

D

GCS 15/15, PEARL

BSL 6.8

E

Temp 36.8

You re-assess the patient 10minutes following admission, and after your initial ABCDE examination.

B

RR36, SaO₂ 93% on continuous nebulisers via oxygen mask, marked accessory muscle use. Nil obvious improvement.

Your Emergency Consultant arrives at the bedside. Together you consider your options for the continued management of this patient.

QUESTIONS

1. Outline the classification system of the severity of asthma on patient presentation? What category does this patient fall into?
2. What are the pharmacological treatment options available for patients with acute severe asthma? Construct a table which outlines the Mechanism of Action, onset, duration, route of administration, dosage, and adverse effects/complications of drugs used in severe asthma, including Salbutamol, Adrenaline, Magnesium, Aminophylline and Hydrocortisone.
3. Outline the assisted ventilation treatments available for patients with acute severe asthma. Include brief discussion of the underlying respiratory pathophysiology in asthma to explain the differences in ventilation strategies for patients with Acute Severe Asthma.
4. Outline your basic ventilator settings for this patient should she deteriorate and require intubation and mechanical ventilation.
5. Outline an appropriate ongoing treatment regime for this patient.
6. Describe the important complications associated with Acute Severe Asthma.

Resources:

- Berston, AD, & Soni, N (2009). *Oh's intensive care manual* (6th ed.). Philadelphia: Elsevier.
- Cameron et al (2009). *Textbook of adult emergency medicine* (3rd ed.). Sydney: Churchill Livingstone.

ANSWERS

1. Outline the classification system of the severity of asthma on patient presentation? What category does this patient fall into?

Table 31.1 Assessment of asthma severity

	Mild	Moderate	Severe
Conscious state	Alert and relaxed	Anxious, difficulty sleeping	Agitated, delirious
Speech	Sentences	Phrases	Words
Accessory muscles	Nil	Mild	Significant sitting upright
Wheeze	Moderate	Loud	Loud or silent
Pulse rate (beats/min)	< 100	100–120	> 120
Peak expiratory flow (% predicted)	> 80%	60–80%	< 60%
P_{aCO_2} (mmHg)	< 45	< 45	> 45

Pulsus paradoxus, when present, indicates severe asthma; however it is an unreliable test.

This patient falls into the Severe Category

What are the pharmacological treatment options available for patients with acute severe asthma? Construct outlining the Mechanism of Action, onset, duration, route of administration, dosage, & adverse effect of drugs used in Severe asthma, including Salbutamol, Adrenaline, Magnesium, Aminophylline, Ketamine & Hydrocortisone.

Medication	Mechanism of Action	Dosage/Route in Severe Asthma	Adverse Effects/Complications
Salbutamol	Beta-2 Agonist	5mg Nebules via oxygen nebuliser continuously May require IV infusion 5-10micrograms/minute	Tachycardia / Arrhythmia Hyper/hypotension Tremor Worsening of V/Q mismatch Hyperglycaemia Hypokalaemia Lactic Acidosis
Atrovent (Ipratropium Bromide)	Anticholinergic Decrease Parasympathetic-mediated bronchomotor tone	500 micrograms every 2-6hours via nebuliser	
Hydrocortisone	Increased B-responsiveness of airway smooth muscle Decreased inflammatory cell response Decreased Mucus secretion	200mg every 6hrs initially (no benefit in >800mg/day) Commence weaning of dose within 1-3 days – change to oral when appropriate	Hyperglycaemia Hypokalaemia Hypertension Acute psychosis Myopathy Increased risk of infection
Magnesium	Blocks calcium channels, possibly blocks Ach release at neuromuscular junction leading to smooth muscle relaxation and bronchodilation	5-10mmol slow IV over 20minutes	Hypotension Flushing Sedation Weakness/Areflexia Respiratory depression Cardiac Arrhythmia
Aminophylline		Loading dose 3mg/kg (if not on oral regular Aminophylline) IV Infusion 0.5mg/kg/hr Reduce dose in liver, cardiac failure, COPD Monitor with drug levels daily	Headache Nausea/Vomiting Restlessness Cardiac arrhythmia Convulsions
Adrenaline	Alpha and beta agonist	Nebulised 5mg every 1-4 hours Infusion 1-20micrograms/min (intubated patients with severe asthma)	Tachycardia/ Arrhythmias Hypertension
Ketamine	Sympathomimetic potentiation and direct effect airway smooth muscle	IV infusion 0.5-2mg/hr for refractory asthma	Increased bronchial secretions Hallucinations

Outline the assisted ventilation treatments available for patients with acute severe asthma. Include brief discussion of the underlying respiratory pathophysiology in asthma to explain the differences in ventilation strategies for patients with Acute Severe Asthma.

Dynamic Hyperinflation (DHI)

Important underlying pathophysiological understanding in severe asthma

Acute severe asthma = severe bronchoconstriction = severe gas trapping in lower airways = dynamic hyperinflation due to incomplete exhalation of gas during normal expiratory time

Ventilation Strategies

Non-invasive ventilation

- externally applied PEEP (EPAP) can help to overcome high intrinsic PEEP (result of gas trapping) and thus reduce the inspiratory threshold work of breathing
- trial of NIV should be considered in patients who are cooperative and can tolerate the face mask, who have moderate to severe dyspnoea or respiratory distress + hypercapnic acidosis + respiratory rate >25 + accessory muscle use
- contraindications include cardiac/respiratory arrest, decreased LOC, severe upper GIT bleeding, haemodynamic instability, facial trauma/surgery, inability to protect airway and clear secretions, or high risk of aspiration
- early trials reporting positive outcomes, but no large randomised trials as yet
- Controversial as to what settings to start on, some advocate for low PEEP (EPAP), others for higher PEEP to offset high intrinsic PEEP. Aim for adequate oxygenation, may need to adopt strategy of permissive hypercarbia initially until bronchoconstriction is relieved.

Invasive Ventilation

- can be life-saving, but also source of significant mortality/morbidity due to inadvertent pulmonary hyperinflation
- necessary in fatigue/increased PaCO₂/decreased LOC
- principle is to avoid excessive DHI and hypoventilation
- Initial settings should be TV 5-7ml/kg, RR 10-12, shortened inspiratory time/long expiratory time (>4sec), PEEP 5
- Hypotension will necessitate decreasing the respiratory rate, in extreme circumstances may need to disconnect ventilator and perform external chest compression to help facilitate expiration of air/gas trapping
- Assess plateau pressures (via inspiratory hold function) and aim for <25cmH₂O
- Assess intrinsic PEEP via expiratory hold function, aim <12cmH₂O
-

Outline your basic ventilator settings for this patient should she deteriorate and require intubation and mechanical ventilation.

Volume Controlled ventilation

- Initial settings should be TV 5-7ml/kg, RR 10-12, shortened inspiratory time/long expiratory time (>4sec), PEEP 5
- Decreased respiratory rate (which will increase expiratory time) if hypotension occurs due to DHI (ensure no pneumothorax as cause)

Outline an appropriate ongoing treatment regime for this patient.
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Optimise Ventilation via Medications

- Continuous salbutamol via nebuliser
- Consider IV salbutamol infusion
- Steroids
- Atrovent
- Potassium replacement/maintenance

Consider Non invasive ventilation

- if worsening/no improvement despite above interventions

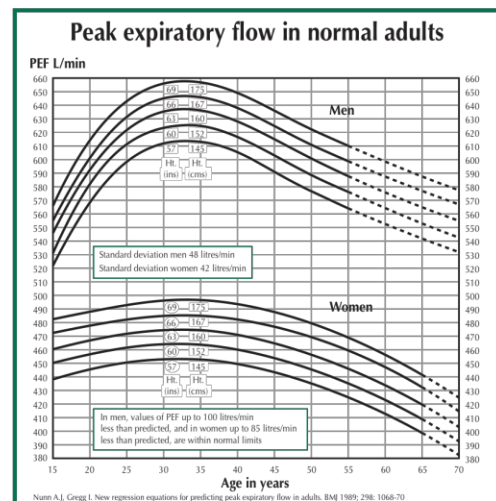
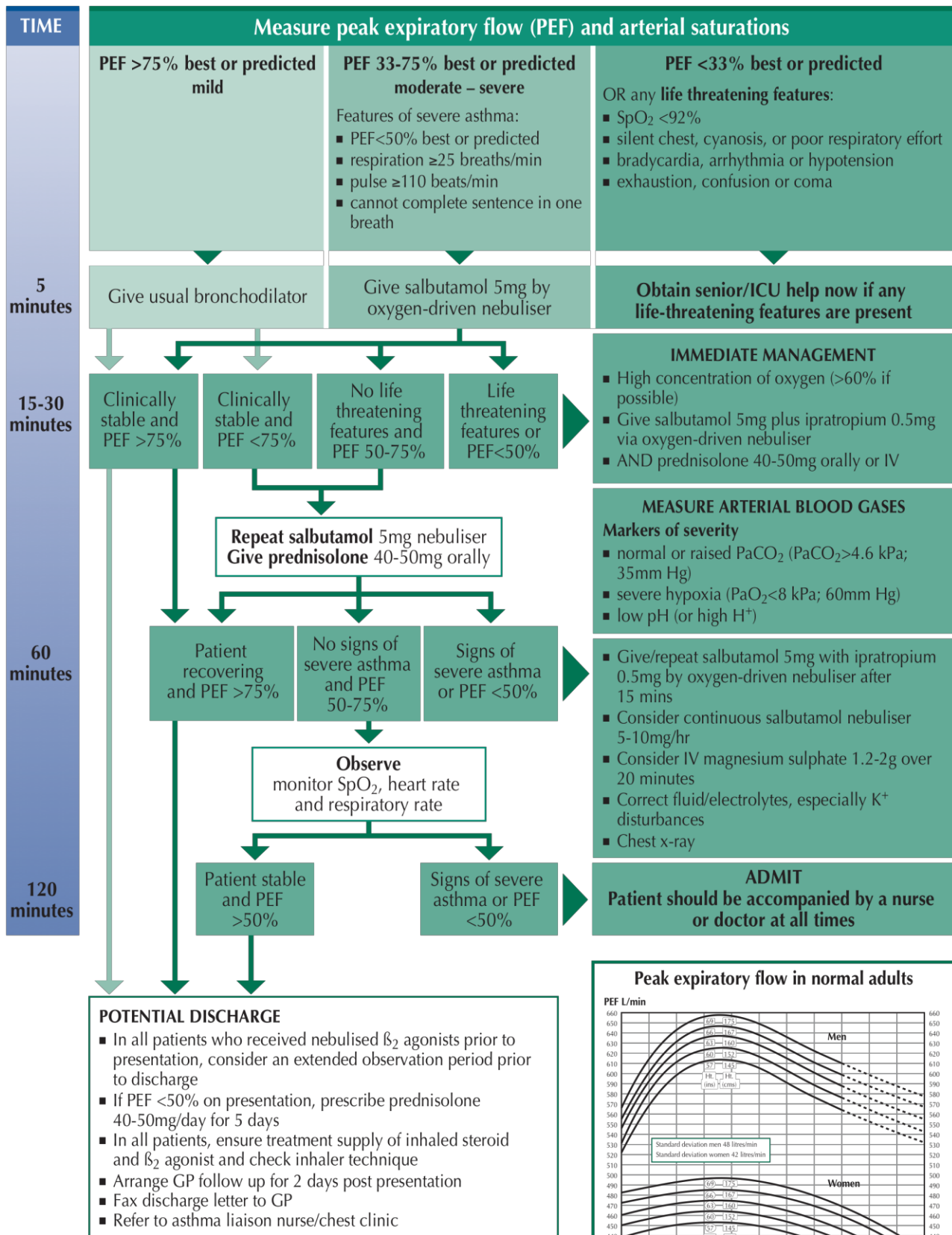
Consider adjunct Medications

- Magnesium
- Aminophylline
- Adrenaline nebulisers

If progresses despite all of the above OR pre-arrest OR arrest

- intubation and mechanical ventilation
- ketamine as induction agent for intubation +/- infusion
- adrenaline infusion

Management of acute severe asthma in adults in A&E



Management of acute severe asthma in adults in hospital

Features of acute severe asthma

- Peak expiratory flow (PEF) 33-50% of best (*use % predicted if recent best unknown*)
- Cannot complete sentences in one breath
- Respirations ≥ 25 breaths/min
- Pulse ≥ 110 beats/min

Life threatening features

- PEF $< 33\%$ of best or predicted
- SpO₂ $< 92\%$
- Silent chest, cyanosis or feeble respiratory effort
- Bradycardia, dysrhythmia or hypotension
- Exhaustion, confusion or coma

If a patient has any life threatening feature, measure arterial blood gases. No other investigations are needed for immediate management

Blood gas markers of a life threatening attack:

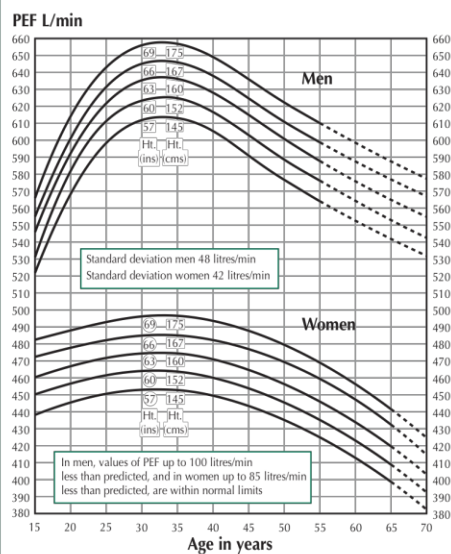
- Normal (4.6-6 kPa, 35-45mm Hg) PaCO₂
- Severe hypoxia: PaO₂ < 8 kPa (60mm Hg) irrespective of treatment with oxygen
- A low pH (or high H⁺)

Caution: patients with severe or life threatening attacks may not be distressed and may not have all these abnormalities. The presence of any should alert the doctor.

Near fatal asthma

- Raised PaCO₂
- Requiring IPPV with raised inflation pressures

Peak expiratory flow in normal adults



IMMEDIATE TREATMENT

- Oxygen 40-60% (CO₂ retention is not usually aggravated by oxygen therapy in asthma)
- Salbutamol 5mg or terbutaline 10mg via an oxygen-driven nebuliser
- Ipratropium bromide 0.5mg via an oxygen-driven nebuliser
- Prednisolone tablets 40-50mg or IV hydrocortisone 100mg or both if very ill
- No sedatives of any kind
- Chest radiograph only if pneumothorax or consolidation are suspected or patient requires IPPV

IF LIFE THREATENING FEATURES ARE PRESENT:

- Discuss with senior clinician and ICU team
- Add IV magnesium sulphate 1.2-2g infusion over 20 minutes (*unless already given*)
- Give nebulised β_2 agonist more frequently e.g. salbutamol 5mg up to every 15-30 minutes or 10mg continuously hourly

SUBSEQUENT MANAGEMENT

IF PATIENT IS IMPROVING continue:

- 40-60% oxygen
- Prednisolone 40-50mg daily or IV hydrocortisone 100mg 6 hourly
- Nebulised β_2 agonist and ipratropium 4-6 hourly

IF PATIENT NOT IMPROVING AFTER 15-30 MINUTES:

- Continue oxygen and steroids
- Give nebulised β_2 agonist more frequently e.g. salbutamol 5mg up to every 15-30 minutes or 10mg continuously hourly
- Continue ipratropium 0.5mg 4-6 hourly until patient is improving

IF PATIENT IS STILL NOT IMPROVING:

- Discuss patient with senior clinician and ICU team
- IV magnesium sulphate 1.2-2g over 20 minutes (*unless already given*)
- Senior clinician may consider use of IV β_2 agonist or IV aminophylline or progression to IPPV

MONITORING

- Repeat measurement of PEF 15-30 minutes after starting treatment
- Oximetry: maintain SpO₂ $> 92\%$
- Repeat blood gas measurements within 2 hours of starting treatment if:
 - initial PaO₂ < 8 kPa (60mm Hg) unless subsequent SpO₂ $> 92\%$
 - PaCO₂ normal or raised
 - patient deteriorates
- Chart PEF before and after giving β_2 agonists and at least 4 times daily throughout hospital stay
- Transfer to ICU accompanied by a doctor prepared to intubate if:
 - Deteriorating PEF, worsening or persisting hypoxia, or hypercapnea
 - Exhaustion, feeble respirations, confusion or drowsiness
 - Coma or respiratory arrest

DISCHARGE

When discharged from hospital, patients should have:

- Been on discharge medication for 24 hours and have had inhaler technique checked and recorded
- PEF $> 75\%$ of best or predicted and PEF diurnal variability $< 25\%$ unless discharge is agreed with respiratory physician
- Treatment with oral and inhaled steroids in addition to bronchodilators
- Own PEF meter and written asthma action plan
- GP follow up arranged within 2 working days
- Follow up appointment in respiratory clinic within 4 weeks
- Patients with severe asthma (indicated by need for admission) and adverse behavioural or psychosocial features are at risk of further severe or fatal attacks
 - Determine reason(s) for exacerbation and admission
 - Send details of admission, discharge and potential best PEF to GP

Describe the important complications associated with Acute Severe Asthma.
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Hypotension / Cardiac Arrest

- due to dynamic hyperinflation resulting in decreased venous return / cardiac output +/- hypovolaemia
- occurs in up to 20% patient who are intubated for acute severe asthma
- enforced apnoea with disconnection of the ventilator for up to 60seconds will usually result in return of spontaneous circulation
- re-commence patient on very low respiratory rate (4-6 breaths/min)

Pneumothorax

- due to DHI and gas trapping
- can progress to tension PTX
- treat with ICC insertion

Acute necrotising Myopathy

- in those who are invasively ventilated and receive neuromuscular blocking agents or very deep sedation with corticosteroids
- may result in need for prolonged ventilation and rehabilitation

CASE FOUR

Short case number: 4_05_04

Category: Ventilation

Discipline: Intensive Care Medicine

Setting: Operating Theatres

Topic: Neonatal Resuscitation_Respiratory Distress Syndrome

CASE



You are on duty as night RMO at a suburban hospital. At 2am you are called urgently to the Delivery Suite to assist the midwives who have just delivered a 32-week child following a very brief premature labour. The obstetrician on-call has not yet arrived, nor has the paediatrician. The baby is 5 minutes old.

He is blue, has a heart rate of 90/min; is floppy; grimaces weakly to a nasal suction catheter; and makes occasional respiratory efforts.

You notice meconium on his face. The hospital has a small maternity unit that manages low-risk cases only.

QUESTIONS

1. What is the baby's Apgar score?
2. Describe assessment of the newborn with particular reference to need for resuscitation.
3. Outline major differences in physiology of the Neonate compared to the adult.
4. Where is this baby best managed? How will you go about getting him to that environment? What advice are you likely to receive from the Newborn Emergency transport Service?
5. The next day, he is very difficult to oxygenate. What are some possible causes? Is there any way that the effects of lung immaturity can be minimised? What is neonatal Respiratory distress Syndrome and how can it be managed?

Resources:

http://www.rch.org.au/nets/handbook/index.cfm?doc_id=447

ANSWERS

What is its Apgar score?

3

Describe assessment of the newborn with particular reference to need for resuscitation.

Need for fetal resuscitation guided by Apgar score 0-10 at 1 and 5mins. 0,1, or 2 points for each of **Heart Rate; Respiratory Effort; Muscle Tone; Reflex Irritability to catheter in nose; Colour**. Apgar 5-7 (mild birth asphyxia) usually respond to stimulation and gentle suctioning of nose, mouth and pharynx. Oxygen sometimes required. Moderate asphyxia (Apgar 3-4) usually respond to bag-mask ventilation with O₂; ABG should be checked and if pH <7.25 HCO₃ administered. Severe asphyxia (Apgar 0-2) needs urgent CPR, intubation and IPPV with O₂, umbi vein catheterisation and HCO₃ 1-2 mEq/kg, IV colloid solution (Albumin) 10mls/kg, Ongoing care in NICU. Call NETS ASAP for advice and transport.

Outline major differences in physiology of the Neonate compared to the adult.

Differences in most areas of physiology. Most obvious is size.

- i) Resp. First few breaths establish FRC. Compliance initially reduced. Initially very small O₂ stores so rapid onset hypoxia. Cannot emphasise strongly enough the benefits of supplemental oxygen in the sick newborn esp prior to intubation etc. Tidal volume similar to adult; higher resp rate, higher VO₂ eg 7ml/kg/min. Fever, restlessness and illness dramatically increase this; PCO₂ rises at twice the rate of an adult. ETT size = (Age/4)+4 mm; Ribs short and horizontal – no “bucket-handle” motion as in adults, so entirely dependent on diaphragm for inspiration. Less compliant lungs, more compliant chest wall – tendency for “indrawing” with any obstruction. More easy fatigue due to less type 1 muscle in diaphragm. Less mature lung – diffusion area relatively smaller than adult.. Premies have surfactant deficiency which accentuates tendency to atelectasis. Overall increased tendency to collapse, shunting and airway closure. Increased work of breathing. All exaggerated in premature neonate.
- ii) CVS. At birth, change from foetal to post-natal circulation. Clamp cord increases SVR; first breaths decrease PVR; pressure changes close Foramen Ovale; PG's close Ductus Arteriosus; A range of insults can lead to reversion to foetal circulation including hypoxia, acidosis, sepsis, hypercarbia. Other differences : newborn heart less capable of increasing contractility, so responds to demands for increases CO by tachycardia. Also less compliant. More tolerant of hypoxia but very intolerant of hypoglycaemia (newborn heart cannot metabolise fatty acids like the older heart). Response to stress tends to be **BRADYCARDIA** rather than tachycardia esp to hypoxia. Neonate normal BP 75/40; HR 125/min; Resp Rate 50/min. Cardiac arrest in the newborn with a normal heart is almost always due to hypoxia, sepsis or hypovolaemia and is usually asystole.
- iii) Fluids and Electrolytes. TBW 80% at term ;60% @ 12/12. Active neonates require twice as much (100mls/kg/day, or 4mls/kg/hour) water as sick neonate. Avoid hypernatraemia or hyponatraemia – the latter usually due to water excess. Avoid hypoglycaemia (decreases cardiac function as well as brain); probably best to use

isotonic “maintenance” fluids especially in situation associated with water retention like post-op and stress situations.

- iv) Immune. Immature. Rely heavily on placentally-transferred IgG. Sepsis
- v) Thermoregulation. Thermoneutral zone for neonate approx 34°C. Hypothermia leads to bradycardia. Poor heat-production – no shivering or behavioural response. Rely on brown fat oxidation.

Where is this baby best managed? How will you go about getting him to that environment? What advice are you likely to receive from the Newborn Emergency transport Service?

The baby needs transfer to a tertiary paediatric hospital with a dedicated Neonatal ICU. This is arranged via NETS. Prior to the arrival of the on-call paediatrician, you will provide bag-valve-mask ventilation AFTER thorough airway suctioning because of the high likelihood of meconium aspiration in this case. As much help as possible should be summoned urgently. An umbilical vein catheter is the easiest venous access; there are 2 veins and one artery in the umbilical stump. 10mls/kg saline or preferably Albumex should be given as well as 1-2 mEq/kg Bicarbonate. Estimate the weight at 2kg for 30 weeks. Intubation with a 2.5 or 3mm ETT after appropriate assistance is available, unless cardiac arrest occurs in which case nothing to lose – just try!. Ensure baby kept warm on resus trolley and check ABG (umbilical artery) and BGL – give iv glucose if low (<??)

What are some possible causes? Is there any way that the effects of lung immaturity can be minimised? What is neonatal Respiratory distress Syndrome and how can it be managed?

4. Respiratory failure in neonates, infants and small children is manifest by tachypnoea (NB normal resp rate of neonate may be 40/min); sternal and rib retraction; accessory use (eg ala nasae flaring); grunting; In young infants, lethargy, apnoea, pallor, bradycardia may be first signs. BEWARE THE BRADYCARDIC INFANT. In the neonate, the effects can be complicated by pulmonary hypertension leading to reversion to a foetal-type circulation. Physical examination in the neonate may demonstrate “normal” breath sounds even in the presence of major pneumothorax or endobronchial intubation; CXR is an essential adjunct.

Causes of respiratory failure in the neonate :

- Upper airway obstruction eg choanal atresia
- Lower airway obstruction eg tracheomalacia, tracheal stenosis, vascular anomalies, Bronchiolar obstruction by MECONIUM,
- Disorders of pulmonary function eg Aspiration esp meconium; Hyaline membrane Disease; Diaphragmatic hernia
- Compression eg Diaphragmatic hernia; pneumothorax
- Neurological failure eg Birth Asphyxia; apnoea of Prematurity

In this baby, the 2 most likely problems would be MECONIUM ASPIRATION SYNDROME and HYALINE MEMBRANE DISEASE

Meconium aspiration occurs in 0.3%; commonest in term or post-dates babies. Combination of obstruction of small airways leading to atelectasis and obstructive emphysema, and a chemical pneumonitis with surfactant abnormalities. Management with O₂ +/- IPPV (beware of barotraumas); Inhaled Surfactant, inhaled NO, and HFO ventilation have improved outcomes in severely affected babies.

HMD is a disease largely of prematurity and intranatal asphyxia. Due to lung immaturity with surfactant deficiency – leads to atelectasis, shunting, increased work of breathing and “wet” lungs. CXR shows “ground glass” appearance. Giving steroids to the mother prenatally if premature delivery is anticipated can hasten surfactant development. The disease is self-limited to 4-5 days but respiratory failure can require high FiO₂, CPAP, HFO ventilation. Inhaled surfactant improves oxygenation and compliance, reduces mortality and pneumothorax risk.